

The reference dataset stores only Z' and Z'' with **no frequency column**, so a frequency-dependent equivalent-circuit (e.g. Randles) fit is not possible. The geometric circle fit recovers R_u directly from the Nyquist arc and needs no frequency data.

Installation

Requires Python 3.10+.

```
pip install -r requirements.txt
```

Dependencies: `numpy`, `scipy`, `pandas`, `openpyxl`, `streamlit`, `plotly`.

Running the app

```
streamlit run app.py
```

Then open the URL shown in the terminal (default `http://localhost:8501`). Select **Use bundled sample** in the sidebar to try it immediately with `sample-data/Book1-original data.xlsx` (current in mA/cm², EIS in Ω ; electrode area 0.04 cm²).

Deploy publicly on Streamlit Community Cloud

This app is ready to deploy for free on [Streamlit Community Cloud](#):

1. Push this repository to GitHub (it already contains `app.py` and `requirements.txt` at the root, which is all the platform needs).
2. Go to **share.streamlit.io** → **Create app**, pick this repo/branch, and set the **main file path** to `app.py`.
3. Click **Deploy**. The platform installs `requirements.txt` and serves the app at `https://<your-app-name>.streamlit.app`.

The sample workbook ships in `sample-data/`, so the deployed app works out of the box; users can also upload their own Excel/CSV files.

Keeping the source code private

Streamlit Community Cloud runs the app **directly from the GitHub repo it is linked to**, so the code must live in that repo — you cannot `.gitignore app.py` and still deploy it (the platform would not have the file). To publish a working app while keeping the source hidden from the public, **make the GitHub repository private**: Community Cloud can deploy from private repos (authorize repo access when creating the app). The deployed app stays publicly reachable at its `*.streamlit.app` URL, while the code is not visible to anyone browsing GitHub. (You can additionally restrict app *viewers* in the app's **Settings** → **Sharing** if you also want to limit who can use it.)

Project layout

Path	Purpose
app.py	Streamlit GUI (EIS/Ru analysis + LSV iR-correction tabs).
ir_compensation/data_io.py	Load EIS / LSV from Excel or CSV (fuzzy column matching).
ir_compensation/eis.py	Circle fit of the Nyquist arc → Ru, Rct.
ir_compensation/correction.py	Apply the iR correction (5–100 % factor; current/Ru unit & area reconciliation).
sample-data/Book1-original data.xlsx	Example data: EIS sheet (Z', Z'' in Ω), LSV sheet (Potential V, Current mA/cm²).

Using the core API without the GUI

```
from ir_compensation import data_io, eis, correction

eis_d = data_io.load_eis("sample-data/Book1-original data.xlsx", sheet=0)
lsv_d = data_io.load_lsv("sample-data/Book1-original data.xlsx", sheet=1)

ru = eis.fit_ru_circle(eis_d.z_real, eis_d.z_imag).ru      # ≈ 27.5 Ω

# Current density (mA/cm²) + Ru in Ω → reconcile with the electrode area.
result = correction.apply_ir_correction(
    lsv_d.potential, lsv_d.current, ru,
    factor_percent=85, current_unit="mA/cm²",
    ru_unit="Ω", area_cm2=0.04,
)
result.potential_corrected    # iR-corrected potentials (V)
result.ru_effective           # Ru reconciled to Ω·cm² (= ru · area)

# Absolute current (mA) + Ru in Ω needs no area:
correction.apply_ir_correction(
    lsv_d.potential, lsv_d.current, ru,
    factor_percent=100, current_unit="mA",    # 100 % = full correction
)
```

Notes

- Unit consistency.** Absolute-current units (A, mA, µA, nA) pair with **Ru in Ω**; current-density units (A/cm², mA/cm², ...) pair with **Ru in Ω·cm²**. The current is scaled to its base SI unit (e.g. mA → A divides by 1000) and, when only one side is area-normalised, the **electrode area** converts Ω ↔ Ω·cm² — so **I·Ru** always resolves to volts. Units are guessed from the column headers and can be overridden in the sidebar.

- **Compensation factor** is selectable from **5 % to 100 %**. 100 % is the full ohmic-drop correction; **85 % is the recommended safe default**, since full positive feedback can over-correct/oscillate when R_u is uncertain. The fold-back guard flags any factor that over-corrects.

Security

Because the app accepts arbitrary uploaded files on a public URL, several safeguards are built in:

- **Upload limits.** `.streamlit/config.toml` caps upload size (15 MB) and message size, keeps XSRF protection on, disables usage telemetry, and hides Python tracebacks from end users (`showErrorDetails = false`).
- **Malicious-file defenses** (`ir_compensation/data_io.py`): Excel archives are checked for **decompression-bomb** expansion before parsing; tables are rejected above row/column caps (memory-exhaustion / DoS guard); the number of parsed column-pairs is capped; and header-derived labels are stripped of control characters / newlines before display or CSV output (prevents CSV/markdown injection).
- **No persisted data.** Uploads are processed in memory only; nothing is written to disk. The sample is **not preloaded**, and a **Clear / reset** button wipes session data.
- **Optional password gate.** Set an `app_password` secret (see `.streamlit/secrets.toml.example`) to require a password; comparison is constant-time. Leave it unset for a fully public app. `secrets.toml` is git-ignored, so credentials are never committed.

To restrict *who* can open the app entirely, you can also limit viewers under the app's **Settings** → **Sharing** on Streamlit Community Cloud.

Citation

If you use this app (including a public Streamlit deployment) in your work, please cite it. Machine-readable metadata is in `CITATION.cff`; a plain-text form:

Kumar, R. (2026). *Automated iR Compensation: EIS fitting and LSV correction* (v1.0.0) [Computer software]. North Carolina Central University. <https://github.com/rajeev4187/LSV-iR-compensation-calculation>

```
@software{kumar_ir_compensation_2026,
  author = {Kumar, Rajeev},
  title  = {Automated iR Compensation: EIS fitting and LSV correction},
  version = {1.0.0},
  year   = {2026},
  url    = {https://github.com/rajeev4187/LSV-iR-compensation-calculation}
}
```

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